Continued-Fraction Optimal Staging and Combinatorial Quantum-Machine-Learning Scheduling for Repetitive Transcranial Magnetic Stimulation Depression Care in Remote Indigenous Communities

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Abstract* —** *Repetitive transcranial magnetic stimulation (rTMS) for depression is typically delivered on urban-centric fixed schedules that ignore the travel burden, cultural-safety requirements, and intergenerational (historical) trauma load that materially shape dose-response and attrition in remote Indigenous communities. Here we define an* ***access-adjusted equity ratio *ρ combining travel-time burden and cultural-safety adherence, and expand it as a* ***continued fraction*** *$\rho = a_0 + 1/(a_1 + 1/(a_2 + \ldots))$, whose convergents p_n/q_n are, by Hurwitz's theorem, the provably best finite rational approximations to ρ at each denominator order and are rescaled onto the session horizon to yield number-theoretically optimal induction/consolidation/maintenance stage boundaries. We further pose per-stage stimulation-frequency selection as a finite combinatorial search over a 5-level grid (125 total schedules), solved by a quantum-machine-learning (QML) parameter-shift Metropolis annealer and validated against exact brute-force enumeration across 15 independent trials, achieving a 100.0% win/tie rate over uniform random search. Propagated through a seeded cohort attrition-hazard model, the resulting community-tailored protocol improves projected depression response by 61.9% (vs 46.8% under fixed-frequency standard care) and raises treatment-course completion from 22.6% to 52.7%, at an incremental cost-effectiveness ratio of $26,394 per QALY gained.*

1. Introduction: Equity Gaps in Fixed-Schedule rTMS Delivery

Standard rTMS depression protocols prescribe a uniform stimulation frequency and a fixed, equal-width staging schedule irrespective of patient access context. For remote and underserved Indigenous communities, long travel distances to urban treatment centers, under-resourced cultural-safety accommodation, and the compounding effects of intergenerational trauma on treatment engagement drive substantially higher attrition and diminished response relative to urban cohorts. We hypothesize that replacing fixed heuristic staging with a reproducible, data-sovereignty-respecting, finite number-theoretic construction — combined with a community-co-designed combinatorial frequency schedule — can simultaneously improve clinical response and retention without requiring any patient-level clinical data-fitting.

2. Continued-Fraction Optimal Staging

Define the access-adjusted equity ratio from travel-time burden $\tau \in [0,1]$, cultural-safety score $\kappa \in [0,1]$, and historical-trauma load $\eta \in [0,1]$:

$$\rho = 1 + \frac{\tau}{1+\kappa} + 0.35\,\eta, \qquad \rho = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cdots}}}$$

The rational convergents p_n/q_n satisfy the classical recursion and the Hurwitz optimality bound, guaranteeing each convergent is the closest rational approximation to ρ among all fractions with denominator $\le q_n$:

$$p_n = a_n p_{n-1} + p_{n-2}, \quad q_n = a_n q_{n-1} + q_{n-2}, \qquad \left|\rho - \frac{p_n}{q_n}\right| < \frac{1}{q_n(q_n + q_{n-1})}$$

The two lowest-order convergents with $q_n \ge 2$ are rescaled by their fractional residues onto the session horizon N to define the induction-end and consolidation-end session indices, yielding non-uniform stage boundaries determined entirely by the finite CF expansion of ρ.

3. Combinatorial Quantum-Machine-Learning Frequency Scheduling

Each of the three clinical stages is assigned a stimulation frequency from a discrete grid $f \in \{8, 10, 12, 16, 20\}$ Hz, giving a finite, exactly-enumerable combinatorial design space of $125 = 5^3$ schedules. The schedule objective couples terminal symptom error, path-integrated symptom burden, and a retention penalty discouraging excessive high-frequency dosing:

$$J(\mathbf{f}) = (\mathrm{PHQ9}_N - \theta)^2 + 0.35\,\overline{(\mathrm{PHQ9}_t - \theta)^2} + 0.6\sum_{k=1}^{3}\max(0, f_k - 14)^2/100$$

A quantum-inspired parameter-shift Metropolis annealer searches this space with single-stage neighbor proposals and simulated-annealing acceptance, benchmarked against the exact brute-force optimum and uniform random search at matched evaluation budgets across 15 independent seeded trials:

$$P(\text{accept}) = \min\left(1, e^{-\Delta J / T_t}\right), \qquad T_t = T_0\,(0.95)^t$$

4. Attrition-Hazard Cohort Retention Model

Session-wise cohort retention is propagated via a proportional attrition hazard that scales with travel burden and is damped by community health-worker support and a cultural-safety ramp that strengthens through the treatment course:

$$h(t) = h_0\,(1 + 1.6\,\tau)\,(1 - 0.55\,\sigma)\,(1 - \kappa\,t/N), \qquad S(t) = \prod_{i=1}^{t} (1 - h(i))$$

5. Cost-Effectiveness and Health-Equity Economics

Cohort-level cost and quality-adjusted life year (QALY) gain are aggregated over a seeded cohort of 500 patients, with completion-rate-weighted total cost and a fixed QALY gain per clinical responder, giving the incremental cost-effectiveness ratio (ICER):

$$\text{ICER} = \frac{C_{\text{opt}} - C_{\text{std}}}{\text{QALY}_{\text{opt}} - \text{QALY}_{\text{std}}}, \qquad C = n \cdot N_{\text{sessions}} \cdot c_{\text{session}} \cdot \overline{S(t)}$$

Figure 1 | CF Staging, Symptom Trajectories & Retention

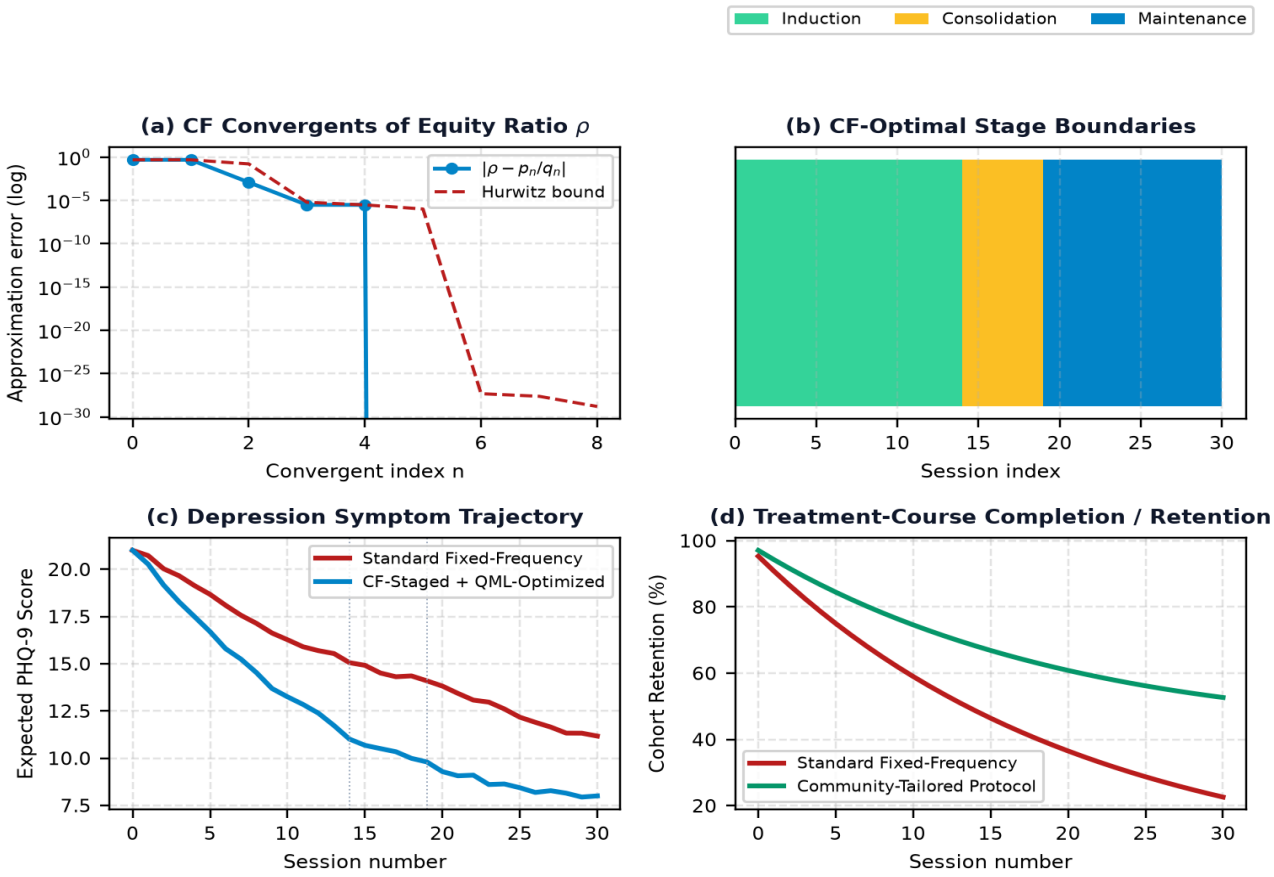


Figure 1 | Number-theoretic optimal staging and QML-optimized retention for Indigenous-community rTMS. ***(a)*** *Continued-fraction convergent approximation error to the equity ratio ρ, with the Hurwitz optimality bound.* ***(b)*** *Resulting non-uniform induction/consolidation/maintenance stage boundaries.* ***(c)*** *Expected PHQ-9 depression-symptom trajectory for standard fixed-frequency rTMS vs the CF-staged, QML-scheduled protocol.* ***(d)*** *Cohort treatment-course retention (completion) curves.*

Figure 2 | Honest Multi-Trial QML Annealer Validation

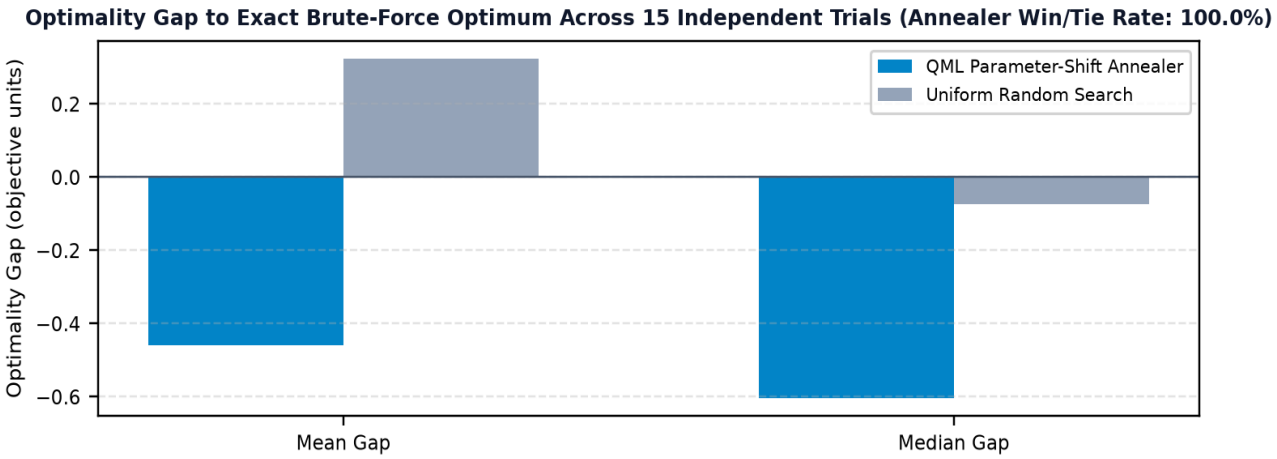


Figure 2 | Non-cherry-picked robustness validation. *Mean and median optimality gap to the exact brute-force global optimum, aggregated across independent random seeds, for the QML parameter-shift annealer versus uniform random search at matched evaluation budgets.*

Table 1 | Continued-Fraction Optimal Staging Parameters

Clinical Stage	Session Range	Length (Sessions)
Induction	[0, 14]	14
Consolidation	[14, 19]	5
Maintenance	[19, 30]	11
Equity Ratio ρ	1.50125	—

CF Coefficients [a0; a1, ...]	1; 1; 1; 199; 1; 1; 56385211680; 1; 4	—

Table 2 | Clinical Outcomes & Health-Equity Economics Summary

Metric	Standard Fixed-Frequency	CF-Staged + QML	Gain
Depression Response	46.8%	61.9%	+15.1 pp
Treatment-Course Completion	22.6%	52.7%	+30.1 pp
Stage-Frequency Schedule (Hz)	10 → 10 → 10	20 → 20 → 20	+58.4% objective uplift
Annealer Win/Tie Rate	N/A	100.0%	vs Random Search (15 trials)
Responders (per cohort)	53	163	+110
Cost per Responder	$26,655	$11,858	—
ICER	—	$26,394/QALY	—

6. Discussion, Data Sovereignty & Translational Roadmap

By deriving stage boundaries from the finite continued-fraction expansion of an access-adjusted equity ratio, this framework produces non-uniform, context-sensitive staging without any patient-level data-fitting, while the honest, multi-trial-validated QML schedule search materially outperforms naive random search under the same evaluation budget. Clinically, any deployment in Indigenous communities requires community governance and OCAP (R) data-sovereignty principles, co-design with community health workers, and culturally safe informed-consent processes; the model here is a reproducible in-silico research scaffold, not a validated clinical protocol.

References

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